

Latent World Models for Robot Planning: Representations, Planners, and Failure Modes

Research Proposal

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August 2026

Abstract

Planning for robots directly from visual observation has converged on a common architecture: an encoder maps images to a latent state, an action-conditioned predictor rolls that state forward, and an optimiser searches over action sequences for the one whose predicted terminal state minimises a distance to a goal. The components of this architecture have been studied largely in isolation—representation objectives in the self-supervised learning literature, trajectory optimisers in the control literature, and end-to-end success rates in the robotics literature—with the result that reported failures are difficult to attribute to a cause.

We propose a research program built around making that attribution tractable. Its instrument is a taxonomy of four failure surfaces—prediction error, proposal coverage, transition realizability, and cost geometry under selection—each carrying an explicit condition that would refute attributing a failure to it (Section 4). Read against the literature on representation objectives and search algorithms (Section 3), the taxonomy exposes the methodological gap the proposed work targets: nearly every existing repair for the latter three surfaces is evaluated under learned dynamics, so its reported gain cannot be separated from ordinary prediction error, and the discriminating experiment—rerunning each repair with candidate dynamics held exact—has, to our knowledge, not been reported for any of them. Section 5 reports a controlled pilot that runs this experiment for one surface: executing every candidate in the simulator, a privileged state-based cost solves a contact-rich task in 64/64 episodes where the standard latent cost solves 0/64 under identical search, and a series of repairs aimed at representation fidelity, run in the same harness, leaves planning unchanged. That result motivates the plan in Section 6: costs trained on the distribution the optimiser itself induces rather than one fixed by the experimenter, a decision-equivalence criterion for how much a representation must preserve, discriminating reruns of the field’s existing repairs under exact dynamics, and an evaluation protocol reporting the quantities—candidate availability, selection regret, proxy–physical rank correlation—that current practice omits.

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1 Introduction

A world model, in the sense used throughout this proposal, is a learned function that predicts the consequences of actions well enough to select among them. The idea is old, but its modern visual form dates to the observation that a compact recurrent latent state suffices to train a controller entirely inside the model’s own imagination (Ha and Schmidhuber, 2018), and to the demonstration that such a latent can be planned in directly from pixels (Hafner et al., 2019). The subsequent years have produced a large and fast-moving literature built on a broadly shared architecture: encode, predict, score, search.

That convergence has an awkward consequence. The architecture is a composition of four blocks, and success requires all four to work, but failure of any one produces the same observable—a task that does not get solved. The encoder may discard the information the task needs. The predictor may produce the wrong future. The search may never propose a good action sequence. The cost may prefer a physically worse sequence to a better one already present in its own candidate population.

End-to-end success rate, the statistic almost universally reported, cannot distinguish these cases. Neither can prediction error, which is measured on a data distribution the planner will not visit.

Scope. This proposal covers latent world models used for *planning* in robot manipulation and navigation: models where control is obtained by optimising over predicted futures at decision time, or by a controller trained against such a model. We treat representation learning and trajectory optimisation as jointly determining behaviour, and we cover the evaluation practices that connect them. We exclude model-free reinforcement learning, video generation evaluated for perceptual quality alone, and policy learning in which no explicit forward model is optimised over.

Research questions. The taxonomy and pilot study below motivate three questions this proposal is organised to answer.

- (1) *Which failure surface binds where?* The diagnostic literature reviewed in Section 3.3 disagrees on this, but the disagreement may be an artifact of protocol—different systems, tasks and dynamics conditions—rather than a real disagreement about mechanism (Section 3.6). Section 6 proposes the discriminating experiment, holding candidate dynamics exact, that would settle this for the field’s existing S3 and S4 repairs.
- (2) *Can a cost or representation be trained on the distribution the planner itself induces, rather than one fixed in advance?* Every objective surveyed in Section 3.1 is fit on a distribution chosen by the experimenter; what a planner consumes is an ordering on a distribution its own search produces after refitting (Equation (1)). Whether an objective can be defined directly on that induced distribution is open, and is the question Section 6 treats as most consequential.
- (3) *Does this failure mode generalise beyond one optimiser and one representation family?* The pilot evidence in Section 5 comes from CEM and two frozen-encoder checkpoints. Section 6 states what would need to hold for the mechanism to be a property of sampling-based search against underdetermined latent costs generally, rather than of this instance.

Approach and preliminary evidence. Answering these questions needs an instrument, and a demonstration that the instrument matters. Both are already in hand. A *design-space organisation* (Section 3) separates what shapes the latent from what searches over it, making visible that these two choices are usually made independently and rarely co-designed. A *failure-surface taxonomy* (Section 4) turns question (1) into an experiment: each surface carries an explicit refutation condition, and applying it to the literature shows that the great majority of proposed repairs are evaluated under learned dynamics, confounding the surface they claim to fix with ordinary prediction error. A *controlled pilot* (Section 5) runs the discriminating experiment for one surface and finds that removing prediction error does not, by itself, close the gap—evidence that motivates the specific directions proposed in Section 6 rather than settling question (1) in general.

Organisation. Section 2 gives background on world models in robot learning—what they are, why the field wants them, how they are used, and why the visual and contact-rich case is hard. Section 3 is the literature review. Section 4 develops the failure taxonomy, Section 5 reports the case study, and Section 6 states the proposed directions.

2 Background

2.1 What a world model is

A world model is an internal, learned model of how an environment responds to action. Given some representation of the current situation and a candidate action, it predicts what the situation will become. The idea long predates deep learning: it is the model in model-based control, the environment model in Sutton’s Dyna architecture, and, in cognitive science, the mental simulation that lets an agent evaluate an action before committing to it.

What distinguishes a world model from a policy is that it is *goal-agnostic*. A policy encodes what to do; a world model encodes what would happen. The distinction has a practical consequence that motivates much of this literature: a policy is tied to the reward it was trained against, whereas a single dynamics model can be reused for any objective one can express over its predictions. This is why world models are attractive precisely where task specifications are numerous, changeable, or unavailable in advance.

2.2 Why world models for robot learning

Four arguments recur, and they are worth separating because different lines of work lean on different ones.

Data efficiency. Robot interaction is expensive in a way that simulated benchmarks are not. Each episode runs in real time, requires resetting the scene, and wears the hardware; collecting the number of transitions that model-free reinforcement learning typically consumes is impractical on physical robots. A world model reuses each observed transition many times, generating training experience or evaluating candidate actions without touching the environment. This is the argument that originally drove the field (Ha and Schmidhuber, 2018; Hafner et al., 2019).

Task generality. The dynamics of a scene are shared across every task performed in it, while rewards are not. A model learned once can serve a new goal at deployment with no retraining, provided the goal can be scored against predicted futures. This argument is strongest for the goal-conditioned formulations that dominate recent visual world-model work, where a new task is specified simply by supplying a new goal image.

Learning without reward labels. Large robot datasets are typically demonstrations or unstructured interaction, carrying no reward annotation. Model-free methods cannot use such data directly. A dynamics model can be fitted to it, and planning then supplies the behaviour that the missing reward would otherwise have shaped (Sobal et al., 2025).

Deliberation before acting. Predicting consequences allows a robot to reject a candidate action before executing it. This matters for safety, and it also permits behaviour that reactive policies obtain only through extensive training—recovering from novel configurations, or composing a sequence no demonstration contained.

2.3 How a world model is used

Three usages appear in robot learning, and the distinction determines what a model must be good at.

As a simulator for policy learning. The model generates synthetic experience on which a policy or value function is trained, in the Dyna tradition; the Dreamer line (Hafner et al., 2019, 2025) is the mature form. Here the model must be accurate on the distribution the improving policy visits, and errors are absorbed by the policy’s own training.

As a substrate for planning at decision time. The model is queried inside an optimisation loop that searches over action sequences at every control step, in the model-predictive control tradition. Here the model is exposed directly to whatever the optimiser chooses to evaluate, including regions no policy would visit. This proposal is concerned with this usage.

As an auxiliary objective. Predicting the future is used as a self-supervised signal to shape a representation, with control obtained by other means. Much recent representation-learning work sits here, and its outputs are frequently adopted by the planning work reviewed later.

2.4 The visual problem

Classical model-based control assumes a compact physical state—positions, velocities, contact forces—identified offline or derived from known physics. It works well where that assumption holds and fails where modern robotics wants to operate. Two obstacles are decisive.

First, *state estimation*. When the only sensor is a camera, the physical state that classical control needs is not directly available, and hand-designed perception pipelines that extract it are brittle to clutter, lighting, and unfamiliar objects.

Second, *model fidelity*. Contact-rich dynamics are stiff and discontinuous, and depend on physical parameters—friction, mass distribution, compliance—that are impractical to identify for every object a robot will encounter.

Learned visual world models address both by declining to estimate physical state at all. Rather than mapping images to positions and velocities and then simulating physics, they learn an arbitrary internal representation and learn how it evolves under action. Ha and Schmidhuber (2018) showed that a controller could be evolved entirely within such a model. Hafner et al. (2019) made the learned representation directly plannable, solving pixel-based continuous control with a sampling optimiser over imagined rollouts, and DreamerV3 (Hafner et al., 2025) showed that one configuration of the recipe transfers across domains without per-task tuning.

The cost of this move is that nothing constrains what the learned state contains. That freedom is the source of both the approach’s generality and the difficulties surveyed in this proposal.

2.5 Pixel-space and latent-space prediction

Visual world models divide by *what* they predict. One family predicts future observations directly, in pixel space, and has benefited from rapid progress in video generation. The other predicts a learned representation of the future and never reconstructs an image.

Two arguments favour latent prediction for control. The first is capacity allocation: a pixel-space objective spends model capacity in proportion to visual variance, and in a robot scene the background, lighting and texture dominate that budget while the task-determining variable often occupies a small fraction of the frame. The second is the unpredictability of detail: much of what a camera records cannot be predicted from the current frame and action, and a model forced to predict it must hedge, producing blurred futures that are poor inputs to a planner. Predicting a representation permits the model to discard what it cannot predict, which is the founding argument for joint-embedding architectures (LeCun, 2022).

Both arguments concern what a representation should *drop*. Neither constrains the geometry of what survives, and Section 4 locates the field’s current difficulty in exactly that gap.

2.6 Planning with a latent world model

The model. Let o_t be an observation and a_t an action. A latent world model comprises an encoder $z_t = E(o_t)$ and an action-conditioned predictor $\hat{z}_{t+1} = f(z_t, a_t)$, applied recursively to obtain an H -step rollout $f^H(z_t, a_{1:H})$. Systems differ in what supervises E and f : pixel reconstruction, latent self-prediction, a downstream value, or nothing at all when E is a frozen pretrained network.

Planning. Given a goal, planning selects

$$a_{1:H}^* = \arg \min_{a_{1:H} \in \mathcal{A}} c\left(f^H(z_t, a_{1:H}), z_g\right), \quad (1)$$

where $\mathcal{A}$ is the set of candidates the optimiser actually materialises and c is a terminal cost. In goal-image formulations $z_g = E(o_g)$ and c is Euclidean distance; in reward-based formulations c is the negative of a learned value. Equation (1) is executed in receding horizon: a prefix of $a_{1:H}^*$ is applied, the state re-encoded, and the optimisation repeated.

Two properties of (1). First, the cost is evaluated on $\mathcal{A}$, a set the optimiser *chooses*, not on the data distribution on which E , f or c were fitted. Second, what the planner consumes is an *ordering* over $\mathcal{A}$; the numerical accuracy of c matters only insofar as it induces the correct ranking. Both properties are routinely elided when models are evaluated by average prediction or decoding error, and both become central to the analysis in Section 4.

2.7 Why this is hard

Three difficulties are intrinsic to the setting and recur throughout the literature reviewed next.

Compounding error. Predictions are applied recursively, so a small per-step error grows over a horizon. This bounds how far ahead any learned model can usefully plan, and it is the difficulty the field understands best.

Partial observability. A single camera frame does not determine the state. Mass, friction, and occluded geometry all affect dynamics and none is visible, so a model must either carry history or accept that its predictions are distributions rather than points.

Contact. Manipulation dynamics are smooth almost everywhere and discontinuous exactly where the task is decided. Near contact, a small difference in action produces a large and qualitatively different outcome, while the visual difference between the two situations immediately beforehand is slight. This is the regime in which reported performance is weakest across the systems reviewed below.

3 Literature review

Section 3.1 covers what shapes the latent, Section 3.2 what searches over it. Two further bodies of work are reviewed separately: the diagnostic literature that has grown up around planning failures (Section 3.3), and the cross-domain literature on optimising against imperfect proxies (Section 3.4), which supplies the statistical vocabulary the robotics work has largely lacked. Section 3.5 covers evaluation methodology, and Section 3.6 synthesises. A substantial part of the work closest to the questions in Section 4 consists of 2026 preprints that have not been refereed; we include them, since excluding recent work would misrepresent the state of the field, but their findings are reported as claims rather than as settled results, and where a comparison rests on an unrefereed source we say so at the point of use.

3.1 Representation objectives

The representation determines what the planner can express and what it can distinguish. We group approaches by the objective that shapes the latent, because that objective—not the architecture—determines what information the latent is permitted to discard.

3.1.1 Reconstruction-based latent dynamics

The founding line trains a latent to support pixel reconstruction and learns dynamics in that latent. Ha and Schmidhuber (2018) compress frames with a variational autoencoder, learn recurrent dynamics in the code space, and evolve a small controller inside the model; the striking result was that a policy trained purely on imagined rollouts transferred back to the environment. Hafner et al. (2019) make the latent directly plannable with a recurrent state-space model combining a deterministic path, which carries information reliably over time, with a stochastic path that models uncertainty; planning uses the cross-entropy method on imagined trajectories, and the system solves continuous control from pixels with far fewer environment steps than model-free alternatives. DreamerV3 (Hafner et al., 2025) extends the line with normalisation and robustness techniques that let one hyperparameter configuration span domains from continuous control to discrete exploration.

Assessment. The reconstruction objective has a well-known liability for control: it allocates capacity in proportion to pixel variance rather than task relevance. In the manipulation setting of Section 2 this is precisely backwards. The Dreamer line partly escapes the criticism because reward prediction shapes the latent alongside reconstruction, which is the same mechanism that Section 3.1.4 identifies as decisive; the purely reconstructive variants do not.

3.1.2 Joint-embedding predictive architectures

Joint-embedding methods predict the *representation* of a target view rather than reconstructing it, discarding unpredictable detail by construction (LeCun, 2022). I-JEPA (Assran et al., 2023) instantiates this for images: a context block predicts the representations of masked target blocks produced by an exponential-moving-average target encoder, with no decoder anywhere in the system. Because the objective admits a trivial constant solution—predict the same vector everywhere—the design of the anti-collapse mechanism is the defining engineering concern of the family. VICReg (Bardes et al., 2022) makes the constraints explicit as a sum of variance, invariance and covariance terms, and LeJEPA (Balestriero and LeCun, 2025) pursues a formulation with fewer heuristics.

V-JEPA 2 (Assran et al., 2025) scales the approach to video and post-trains an action-conditioned predictor on robot interaction data, planning zero-shot with a sampling optimiser against latent distance; it remains the principal demonstration that this family can drive a physical robot. V-JEPA 2.1 (Mur-Labadia et al., 2026) reports that a dense predictive loss and deeper self-supervision substantially improve contact-rich grasping over that baseline. That result matters beyond its own contribution: it is direct evidence that spatial feature density, and not only the planning objective, governs contact performance, and Section 6 treats it as the leading competing explanation for the findings in Section 5. LeWorldModel (Maes et al., 2026) trains encoder and predictor jointly from pixels, dispensing with the frozen-encoder assumption that most planning work inherits.

Assessment. Two structural facts about this family recur throughout the rest of this review. There is no decoder, so a latent distance cannot be inspected to determine what it measures—any claim about what the cost is doing requires building and separately validating a probe. And the target encoder observes the true future, which admits solutions in which the predictor matches its target without making full use of the action input; this is taken up in Section 3.3.3.

3.1.3 Frozen pretrained visual features

A third strategy avoids representation learning for control altogether: freeze a strong self-supervised encoder and train only the dynamics. DINO-WM (Zhou et al., 2025) freezes DINOv2 (Oquab et al., 2024) patch tokens, trains an action-conditioned predictor over them, and plans zero-shot against latent L_2 distance to a goal image, with goals specified visually rather than as states or rewards. The appeal is modularity and sample efficiency: one visual backbone serves every task, only the dynamics must be relearned, and no reward engineering is required.

Terver et al. (2026) provide the most comprehensive controlled comparison available for this family, systematically ablating design choices—encoder frozen versus trained, action-conditioning mechanism, rollout length, cost form—and releasing checkpoints and a planning stack across several benchmarks. Their finding that Euclidean latent distance with sampling-based search is the effective default is why L_2 is the object of study in most subsequent diagnostic work rather than an arbitrary choice. Sobal et al. (2025) extend latent-dynamics planning to reward-free offline data, arguing that planning is the right way to exploit data with no reward labels, and Bar et al. (2024) apply the formulation to navigation at scale.

Assessment. The unexamined assumption is that a representation optimised for generic visual prediction induces a geometry in which distance is a usable planning cost. The pretraining objective never saw a planner, a goal, or an action. Section 3.3.4 reviews the accumulating evidence that this assumption fails, and it fails hardest in exactly the contact regime described in Section 2.

3.1.4 Value- and task-aware latents

TD-MPC2 (Hansen et al., 2024) occupies the opposite end of this axis. The latent is trained jointly with reward and value prediction and is never required to reconstruct anything; planning uses MPPI with a learned terminal value that substitutes for the tail of the horizon. The system scales to a single agent spanning many tasks and embodiments. The line traces back to MuZero (Schrittwieser et al., 2020), which trains a latent purely to support its own reward, value and policy predictions—never grounding it in observations or an explicit dynamics-reconstruction loss at all—and searches over it with Monte Carlo tree search rather than the sampling-based MPC that dominates Table 2; TD-MPC2 keeps the value-trained latent but swaps tree search for MPPI, which is the more practical choice in continuous action spaces.

Assessment. This is the clearest existing demonstration that cost geometry is a property one can *train for* rather than an inevitable consequence of latent prediction, and it is the natural comparison for any claim that planning failure is intrinsic to this class of models. Its limitation is the requirement for a reward signal, which is precisely what goal-image formulations are designed to avoid; whether an equivalent can be obtained reward-free is one of the central open questions of Section 6.

3.1.5 Explicitly structured latents

A recent group imposes structure that the generic objectives do not. Kim (2026) frames the problem as latent state design under sufficiency constraints, organising methods by what their state is *for* rather than by architecture, and arguing that different downstream uses—prediction, control, memory, counterfactual reasoning—impose different and sometimes incompatible sufficiency requirements. Wang et al. (2026) regularise the curvature of latent trajectories, motivated by the perceptual straightening hypothesis, so that Euclidean latent distance better approximates geodesic distance; they report improved conditioning of the planning objective and higher success on goal-reaching tasks, which is a direct intervention on the geometry rather than on the cost.

Table 1: Representation objectives and their consequences for the planning cost.

Family	Representative work	What it discards	Consequence for the cost
Reconstruction	World Models, PlaNet, Dreamer	little; capacity follows pixel variance	cost inherits irrelevant visual structure
Joint-embedding	I-JEPA, V-JEPA 2, LeWorldModel	unpredictable detail	no decoder; latent distance uninterpretable
Frozen pretrained	DINO-WM, JEPA-WM study, PLDM	whatever pretraining discarded	geometry fixed before the task is known
Value-aware	TD-MPC2	anything the value ignores	cost is trained, not assumed; requires reward
Structured	straightening, sufficiency, object-centric	by explicit constraint	targets the interface directly

Object-centric and invariance-based variants (Nam et al., 2026; Toso et al., 2026) constrain the latent toward factors the task depends on.

Assessment. This group is the most directly responsive to the diagnosis in Section 4, because it targets the representation–planner interface explicitly rather than treating representation quality as a proxy for it. Its evidence base is currently thin and largely on simpler benchmarks.

3.2 Planning and control algorithms

3.2.1 Sampling-based model-predictive control

The dominant choice is the cross-entropy method (Rubinstein, 1997): sample a population of action sequences from a parametric distribution, evaluate each under the model, retain an elite fraction, refit the distribution to those elites, and iterate. It requires no gradients, tolerates discontinuous costs, and parallelises trivially, which is why it appears in nearly every system reviewed above. iCEM (Pinneri et al., 2020) adds the refinements that make it practical in real time—colored-noise sampling that produces temporally correlated action sequences rather than white noise, momentum on the distribution parameters, and warm-starting from the previous solution—reporting order-of-magnitude reductions in the sample budget needed for real-time control. MPPI (Williams et al., 2017) replaces elite truncation with an exponential weighting derived from a path-integral formulation of stochastic optimal control.

Assessment. The refitting loop deserves emphasis, since it is the mechanism behind several results in Section 3.3.4. These optimisers do not sample independently: they concentrate the candidate distribution wherever the cost reads low. If the region where the cost reads low is a region where the cost is *wrong*, the optimiser moves its own evaluation set into that region and keeps it there. The distribution on which the cost is queried is therefore determined jointly by the cost’s errors and by the search, and is not the distribution on which any component was validated.

3.2.2 Gradient-based planning

Because f and c are differentiable, one can descend Equation (1) directly in action space. This is far cheaper per iteration than sampling, but queries the model in a mode it was never trained for—through input gradients rather than forward prediction—and reliably produces out-of-distribution or adversarial action sequences. Parthasarathy et al. (2025) characterise this train–test gap explicitly and repair it by synthesising training data that smooths the induced planning landscape; Wang et al. (2026) attack the same problem from the representation side by improving the conditioning of the objective.

Table 2: Planning algorithms and their characteristic failure exposure.

Planner	Representative work	Principal exposure
CEM / iCEM	Rubinstein (1997); Pinneri et al. (2020)	refitting concentrates candidates where the cost reads low, including where it is wrong
MPPI	Williams et al. (2017)	as above, with soft weighting instead of truncation
Gradient descent	Parthasarathy et al. (2025)	queries the model through input gradients, a mode it was not trained for
Amortised (GC-IDM)	Nguyen et al. (2026)	no search to exploit the cost, but inherits the geometry of the latent it was fitted on
Hierarchical	Zhang et al. (2026)	shorter horizon per level; the terminal cost is still queried at every level

3.2.3 Amortised and search-free control

Search at decision time is expensive, and Nguyen et al. (2026) ask whether it is necessary. They replace it with a goal-conditioned inverse dynamics model mapping current latent, goal latent and remaining horizon directly to an action, matching or exceeding CEM in seven of eight settings at two orders of magnitude lower per-decision cost. Their argument is that under a sufficiently regular latent geometry—their study uses a model whose latent is explicitly regularised for smoothness and uniformity—much of what search recovers is already locally encoded, so the search is redundant.

Assessment. The study also sweeps CEM, MPPI, iCEM and gradient planning, making it the clearest available caution against drawing conclusions about latent planning from a single optimiser. Its positive result is conditional on the regularised latent it studies, so it does not establish that amortisation helps for arbitrary representations.

3.2.4 Hierarchical decomposition

Long-horizon tasks strain a flat optimiser, whose candidate space grows exponentially in horizon. Zhang et al. (2026) plan hierarchically with latent world models, decomposing a goal into subgoals so that each level faces a shorter horizon. The relevance to Section 4 is indirect but important: a shorter horizon leaves a cost less room to be exploited, making hierarchy a plausible mitigation for cost-geometry failures as well as a scalability mechanism.

3.3 Diagnostic and failure-analysis work

A distinct body of work, most of it from 2025–2026, studies why these systems fail rather than proposing new ones. It is the most directly relevant literature to this proposal and the least settled.

3.3.1 Dynamics error and its exploitation

Janner et al. (2019) establish the classical result: a policy optimised inside a learned model exploits that model’s error, and the damage can be bounded by limiting rollout length according to model confidence. Lambert et al. (2020) sharpen the point into the form most relevant here, showing that the objective used to train a dynamics model is not monotonically related to control performance—a model that is better by likelihood can control worse. Bhamidipaty et al. (2026) state the general case for planning against imperfect models. The offline setting produced the sharpest version of the same fix: MOPO (Yu et al., 2020b) and MOREL (Kidambi et al., 2020) penalise a candidate by an estimate of the model’s own uncertainty at that candidate, so the optimiser is pushed away from

exactly the region it would otherwise exploit, rather than being stopped early as in the rollout-length repair. This surface is comparatively well understood, and its remedies are standard.

3.3.2 Proposal coverage

Gao et al. (2026) make a case that is easy to overlook and hard to dismiss. Replacing the learned predictor with literal environment rollout while holding encoder, cost, budget and replanning cadence fixed, they find that on their failing cells the candidate population contains almost no goal-reaching sequence at all—so no improvement to the cost could have helped. Their remedies are correspondingly proposal-side: demonstration retrieval to initialise the search, a start-goal-action compatibility score, and a reliability gate that modulates how far the planner trusts the model.

Assessment. This is the principal competing explanation for any claimed cost-geometry failure, and distinguishing the two requires measuring coverage and selection on the same population. Their substitution of environment rollout for the learned predictor also establishes priority for the oracle-dynamics construction used in Section 5.

3.3.3 Transition realizability and action grounding

A rollout may end near the goal while passing through transitions that no sequence of actions could produce. Seo et al. (2026) add per-step inverse-dynamics cycle residuals to the planning cost with an adaptive scale, rejecting candidates that the terminal cost alone would prefer, and report improvements across four action-conditioned world models and six manipulation and navigation tasks. Liu et al. (2026) decompose verification into state plausibility and action reachability using a sparse inverse model and cycle consistency, using the resulting verifier for self-improvement in under-explored action regimes. Chen (2026) build probes on real encoded versus model-predicted transitions and summarise action-semantic preservation in a transfer matrix, collapsible into a screening score that ranks checkpoints without planner-coupled evaluation—reporting order-of-magnitude speedups over simulator rollout for model selection.

3.3.4 Cost geometry and selection

The newest strand, and the one on which the literature has converged most rapidly. Li et al. (2026b) observe that a candidate can be close in latent space yet unreachable within a finite horizon—latent proximity does not imply dynamical proximity—and add a budget-conditioned reachability head trained from trajectory offsets and temporal hard negatives.

Li et al. (2026a) conduct the most direct study available. Holding encoder, dynamics, sampler, optimiser, budget and evaluation manifest fixed and replacing only the terminal cost, they report: a raw latent cost that misranks candidates despite nearly perfect linear decodability of task coordinates from the same latent; a true task-state cost reaching full success under identical settings; a solver-stress control in which substantially more search fails to rescue the wrong objective; same-candidate selection audits that compare cost orderings on identical candidate sets; and a subspace intervention showing that the coordinate rowspace carrying most control utility accounts for under one percent of raw latent error. They also deploy their repaired metric inside full CEM rather than evaluating it offline.

Assessment. These two works establish that the phenomenon is real and not an artifact of one system. It is worth stating explicitly what Li et al. (2026a) already covers, since it bounds what any subsequent work can claim: candidate-level selection auditing, a state-based positive control, and deployment of a repaired cost under search are all present. What is absent from this strand is a study conducted with candidate dynamics held exact, which is the gap Section 4 identifies.

3.4 Optimisation against imperfect proxies

The cost-geometry strand has a precise analogue outside robotics, and importing it correctly matters for interpreting results.

Smith and Winkler (2006) prove the foundational result, in a decision-analysis setting: selecting the extremum over noisy estimates guarantees that the selected item’s true value is worse than its estimate, *even when every estimate is unbiased*. The consequence for this literature is methodological and, so far, largely unobserved: a demonstration that a proxy cost misranks candidates establishes nothing unless it exceeds a matched-noise baseline, because pure selection over unbiased noise already produces misranking.

Skalse et al. (2022) formalise when improving a proxy degrades true performance, showing that the conditions under which a proxy is “unhackable” are restrictive enough to be practically unattainable. Gao et al. (2023) measure the relationship empirically at scale, tracking gold reward against proxy reward as optimisation pressure increases under both reinforcement learning and best-of- n selection, and documenting the characteristic rise-then-fall; that curve shape is what licenses the term *overoptimisation*, and its absence should preclude the term. Karwowski et al. (2024) give a measurable definition of optimisation pressure and study early stopping as a mitigation. The same shape has been documented for inference-time selection in language models (Khalaf et al., 2025; Helff et al., 2026), where verifier-guided search degrades true quality past a threshold of search effort.

Assessment. This literature supplies the vocabulary—proxy, reference, optimisation pressure—and one non-negotiable control, the matched-noise null. What it lacks, and what the robotics setting uniquely offers, is physical ground truth for the selected candidate: in language-model work the “gold” signal is itself a learned or human judgement, whereas a simulator can report exactly where the object ended up.

3.5 Evaluation methodology and benchmarks

MetaWorld (Yu et al., 2020a) supplies most contact-rich manipulation evaluation in this literature. Its value here is a scriptable success criterion with an explicit tolerance, which permits the physical ground truth Section 5 depends on. Navigation and 2D benchmarks recur as sanity checks, but several are saturated for modern models and carry little discriminative power. Video-model benchmarks have begun to test physical adherence and action-following directly (Yang et al., 2026), reporting that visual fidelity is a poor proxy for action fidelity and that scale does not reliably improve action following. Paired intervention protocols evaluate causal response by changing one physical variable at a time (Cai et al., 2026), with Xia et al. (2023) supplying the underlying formalism for counterfactual identification. Hou et al. (2026) provides a wider taxonomy of the robot-learning world-model landscape.

Yu et al. (2026) argue from a position paper that predictive quality should be connected to downstream decision quality rather than reported alone. Workshop authors (2026) identify a protocol confound with direct consequences for comparability: terminal-at-horizon scoring can be misaligned with prefix execution under receding-horizon control, so planning horizon, executed prefix, terminal versus running cost, and replanning cadence all change the conclusion.

Assessment. Three quantities are essentially never reported and each is cheap to compute given an existing planning stack: whether a physically successful candidate was present in the population at all; the physical regret of the selected candidate relative to the best available one; and the rank correlation between proxy and physical outcome on identical candidates, measured separately at the first and last search iteration. Success definitions also differ materially across papers—an any-step

Table 3: Four failure surfaces, with refutation conditions.

Surface	Observable signature	What refutes an attribution to it
S1. Prediction error	A physically good action sequence is rolled out incorrectly	Failure persists when candidate dynamics are made exact
S2. Proposal coverage	No physically good candidate ever enters the population	Good candidates are demonstrably present and are not selected
S3. Transition realizability	Predicted intermediate transitions are inconsistent with the actions conditioning them	Failure persists when transitions are exact by construction
S4. Cost geometry / selection	Candidates with low proxy cost have poor physical outcome; the gap persists or widens with search	Proxy and physical rankings agree on the optimiser’s own realised populations

latch and a strict episode-end flag can differ by a large factor on identical runs, enough to reverse a comparison between systems—and Chen (2026) additionally warn that diagnostic scores are not comparable across model families without calibration, since each model defines its own latent scale.

3.6 Synthesis

Three observations close the review.

Representation and planner are chosen independently. Almost every system reviewed pairs a representation objective selected on self-supervised grounds with an optimiser selected on control grounds, and the interface between them—the cost—is assumed rather than designed. The exceptions are instructive: TD-MPC2 trains the latent against the decision, and the structured-latent group shapes the geometry deliberately. Both report that the interface improves when it is treated as a design target.

The diagnostic literature disagrees, and the disagreement is unresolved. IMWM concludes proposal coverage; TRM concludes terminal-metric misranking; ACID concludes transition realizability. These conclusions are drawn on different systems, different tasks, and different protocols, so they are not directly comparable, and the field currently has no way to say which failure binds where.

The statistical controls are largely absent. The proxy-optimisation literature has established for two decades that selection over noisy estimates degrades true value by itself. Almost no work in latent planning reports the matched-noise null that this implies, which means the quantitative magnitude of reported misranking effects is difficult to interpret.

Section 4 proposes a taxonomy intended to make the second of these tractable.

4 A taxonomy of failure surfaces

Reported failures in this literature are attributed inconsistently, and the attributions are rarely tested. We propose the taxonomy in Table 3, whose distinguishing feature is the third column: for each surface, the observation that would *refute* attributing a failure to it. A taxonomy without refutation conditions organises vocabulary; one with them organises experiments.

Table 4: Failure-surface matrix: which surface each work’s evidence primarily targets, and under what dynamics condition that evidence was collected. S1 prediction error, S2 proposal coverage, S3 transition realizability, S4 cost geometry/selection — full definitions and refutation conditions are in Table 3. • primary claim, ◦ secondary claim.

Work	S1 pred. err.	S2 coverage	S3 realiz.	S4 cost geom.	Dynamics	Contribution type
MBPO (Janner et al., 2019)	•				learned	repair (rollout limit)
Objective mismatch (Lambert et al., 2020)	•			◦	learned	diagnosis
Train–test gap (Parthasarathy et al., 2025)	•				learned	repair (train-time synthesis)
Imperfect WMs exploitable (Bhamidipaty et al., 2026)	•			◦	learned	diagnosis
IMWM (Gao et al., 2026)		•		◦	exact	repair (retrieval, gate)
Latent Geometry (Nguyen et al., 2026)		•			learned	repair (amortisation)
ACID (Seo et al., 2026)			•	◦	learned	repair (cost term)
World Action Verifier (Liu et al., 2026)			•		learned	repair (verifier)
ATM (Chen, 2026)			•		learned	diagnosis + objective
TRM (Li et al., 2026a)		◦		•	learned	repair (terminal metric)
RC-aux (Li et al., 2026b)			◦	•	learned	repair (reachability)
Temporal straightening (Wang et al., 2026)				•	learned	repair (representation)
Optimizer’s curse (Smith and Winkler, 2006)				•	n/a	theory (null model)
Goodhart / overoptimisation (Skalse et al., 2022; Gao et al., 2023; Karwowski et al., 2024)				•	n/a	theory + measurement
§5 case study (this proposal)	—	◦	—	•	exact	diagnosis

The surfaces map onto the review as follows: S1 to Section 3.3, S2 to Gao et al. (2026), S3 to the action-grounding group of Section 3.3.3, and S4 to Section 3.3.4 with the statistical machinery of Section 3.4. One asymmetry is worth noting: S1 is the only surface that can be closed *by construction*, by supplying exact dynamics, which makes it the natural control condition for studying the other three.

4.1 Applying the taxonomy: a methodological gap

Table 4 places the work that makes a failure claim against the four surfaces. The column to read carefully is *Dynamics*, which records the condition under which each paper’s own evidence was collected.

Two observations follow, and they are the analytical core of this proposal.

First, nearly every proposed repair is evaluated under learned dynamics. By the refutation conditions of Table 3, a gain measured in that condition cannot be attributed to S2, S3 or S4 rather

than to S1: improving any component correlated with prediction quality will improve outcomes whenever prediction is the binding constraint. The discriminating experiment—running the same repair with exact candidate dynamics, where S1 is closed by construction—has to our knowledge not been reported for any of the S3 or S4 repairs surveyed above. This is a tractable and high-value gap.

Second, the field’s diagnoses appear to conflict but need not. The disagreement noted in Section 3.6 dissolves if these are read not as mutually exclusive claims about latent planning in general, but as claims about which surface binds in a particular regime. The useful question is therefore not which diagnosis is correct but which binds where—a question the taxonomy makes askable and the current evidence base cannot answer, because the studies share neither a protocol nor a dynamics condition.

5 A controlled case study in cost attribution

To make the preceding argument concrete, we summarise a study designed around the refutation conditions of Table 3. It is included as an instance of the methodology this proposal advocates rather than as a general result; its scope is two released checkpoints and two contact-rich manipulation tasks.

Setting. The tasks are tabletop manipulation with a simulated Franka arm, the setting in which the contact difficulty of Section 2 is most acute. Two properties of it are load-bearing for what follows. Success is declared at object-to-goal distances of a few centimetres, so a representation that localises an object to within ten centimetres is useless for the task while appearing accurate by most reported metrics. And free-space reaching in the same suite is comparatively forgiving, so success rates pooled across a task suite can conceal complete failure on the contact subset—which is why the study reports contact and free-space cells separately.

Design. The study closes S1 by construction. At every replanning step the simulator state is snapshotted; each candidate action sequence is executed in the simulator, its terminal observation rendered, and that observation encoded by the frozen encoder. Candidate dynamics are therefore exact while the representation and the terminal scorer remain the deployed ones. Everything else—encoder, optimiser, budget, horizon, success criterion—is held fixed across arms, so only the cost varies. Three costs are compared: the released latent L_2 ; a supervised readout of object and hand position from frozen tokens, composed into the same functional form as the reference; and a reference computed from privileged simulator state, which serves as a positive control for the search budget and is not a deployable method. Evaluation uses the simulator’s task flag at the final environment step on MetaWorld tasks (Yu et al., 2020a), with checkpoints from Terver et al. (2026).

Result. Table 5 reports the comparison. Under identical search, the state-based reference solves push in 64/64 episodes and pick-place in 49/64, while the latent cost used by the released systems solves 0/64 in every cell. The supervised readout reaches at most 5/64 despite held-out object decoding error of 1.64–1.71 cm, comfortably inside the 5 cm success radius. Since prediction error is zero by construction and the planner demonstrably suffices, the binding constraint in this setting is the mapping from representation to cost.

Mechanism. Why an apparently accurate readout fails is answered only by conditioning on the optimiser. Agreement between the readout cost and the physical reference is moderate on initial candidate populations (Spearman 0.43–0.55; reference top-10 recall 0.19–0.26) and collapses on

Table 5: Terminal-cost comparison under exact candidate dynamics; 64 paired environment seeds, strict episode-end success. The reference row is a privileged positive control for the search budget, not a deployable method.

Cost	DINO push	DINO pick	JEPA push	JEPA pick
Simulator-state reference	64/64	49/64	64/64	49/64
Latent L_2	0/64	0/64	0/64	0/64
Supervised state readout	5/64	0/64	1/64	1/64

Table 6: Interventions in the same harness. Every row improved its target quantity; none moved contact planning outside the 0–2/16 baseline band.

Intervention	Target quantity improved	Planning	Reading
Post-hoc cost variants (learned metric, decoded object, composed readout)	various; all match or beat L_2 offline	0–2/16	no post-hoc cost on the frozen encoder crossed
Off-policy-robust readout	off-policy decode 78 $\rightarrow$ 91.5% < 5 cm	1/16	readout precision was not the constraint
Relearned representation adapter	decode median 6.9 cm	1/16	nor was the readout’s parameterisation
Encoder fine-tuning (LoRA), 5 seeds	grounding 94–100% < 5 cm	{5, 0, 2, 1, 1}/16	inside the frozen band; the 5 is seed variance
Ensemble disagreement penalty	penalty demonstrably active across λ	{2, 1, 2, 1}/16	members share the backbone’s blind spots
Amortised search-free control	removes the optimiser entirely	null	removing search is not sufficient
Grounded dynamics channel	counterfactual tracking corr +0.68/+0.70; cost surface +0.089 [+0.022, +0.162]	0 episodes flipped	grounding improved, planning unchanged
Selection-aware encoder objective, 4 arms $\times$ 3 seeds	validation selection regret 0.0020–0.0069	0–5/16	only the ranking-based arm was non-degenerate

the populations the optimiser itself produces after refitting (0.11–0.16; 0.07–0.10). On identical populations the readout’s selection incurs 2.05–2.48 cm of within-population regret against the reference, versus 1.07–1.15 cm for a residual-matched permutation null—so the excess of 0.91–1.36 cm exceeds the optimizer’s-curse baseline of Smith and Winkler (2006), with all confidence intervals above zero. Intervening on the refitting cost alone degrades the best candidate subsequently available by 1.51–2.01 cm, in the same direction in all eight seeds. Exact-success candidates remain sparse in final populations (8.0%, 3.6%, 0%, 0% across cells), which is a partial concession to the coverage account of Gao et al. (2026): availability is measured after prior refits, so it does not independently establish a proposal failure.

Repairs attempted. Table 6 summarises interventions run in the same harness. The pattern is more informative than any single row: each improved the quantity it targeted, and none moved contact planning outside the baseline band. Raising off-policy readout accuracy from 78% to 91.5% within 5 cm transferred no planning success; restoring counterfactual action tracking to a correlation of 0.68–0.70 flipped no episodes; light encoder fine-tuning reaching 94–100% grounding accuracy stayed inside the frozen band across five seeds; and an ensemble-disagreement penalty, the standard model-based remedy, was flat because ensemble members share the frozen backbone’s blind spots.

Scope. Two checkpoints from one design lineage, one simulator, one task family; uncertainty is over environment seeds, not over checkpoint training seeds. The study does not establish that latent costs fail in general, nor that scale cannot help—indeed Mur-Labadia et al. (2026) is direct evidence that denser features improve contact performance, and running a dense-feature checkpoint through the same protocol is the most informative single extension available.

6 Proposed research directions

Costs trained on the distribution the planner induces. The recurring pattern of Table 6—fidelity improves, planning does not—has a structural reading. Every objective in Section 3.1 optimises how well some learned quantity matches a physical one on a distribution chosen by the experimenter. What a planner consumes is an ordering on a distribution chosen by the *optimiser*, after refitting has moved it. Objectives defined on rankings within realised populations, rather than on values over a fixed dataset, are largely unexplored; the limited evidence available suggests scalar regression onto true cost is not the right form, while group-relative ranking objectives are more promising and essentially untested at scale.

Decision equivalence as a representation criterion. Kim (2026) frames representation design as a question of sufficiency: what must the state preserve for its intended use? For planning, the natural criterion is preservation of the argmin over a family of goals, which is far weaker than reconstructing the state and far stronger than predicting the next latent. TD-MPC2 (Hansen et al., 2024) achieves something like it through value supervision; whether an equivalent is obtainable in the reward-free, goal-image setting is open, and would be the most direct response to the evidence in Section 5.

Discriminating experiments for existing repairs. The gap identified in Table 4 is directly actionable. Running the S3 and S4 repairs—inverse-consistency costs, reachability heads, learned terminal metrics—under exact candidate dynamics would establish, for each, whether its gain survives when prediction error is removed. A repair that loses most of its benefit was repairing S1; one that survives is a genuine counterexample to the attribution in Section 5. Either outcome is informative, and the experiment is inexpensive relative to its value. The prerequisite is a positive control reproducing the original gain under learned dynamics, without which a null result is uninterpretable.

Whether hierarchy changes the picture. Amortisation removes the optimiser but inherits the latent geometry, and the one controlled test available found it insufficient. Hierarchical decomposition (Zhang et al., 2026) differs in kind: it shortens the horizon each cost must span, leaving less room for a candidate to reach a low-cost region that is physically wrong. Whether this helps, or merely relocates the same failure to every level, is unresolved and directly testable.

Representation quality as the competing explanation. The strongest alternative to a cost-geometry account is that the frozen features themselves lack the spatial precision contact requires. Mur-Labadia et al. (2026) report large contact-task gains from denser predictive features, which is exactly what that account predicts. No published study runs a dense-feature checkpoint through a cost-attribution protocol, so the two explanations remain entangled. Resolving this should be a priority, since they imply opposite research programmes—improve the encoder, or change the objective that shapes it.

Generality across optimisers. Almost all evidence about cost-geometry failure comes from CEM. Since the proposed mechanism depends on distribution refitting, it should transfer to MPPI and iCEM and behave differently for gradient and amortised controllers, but that is a prediction rather than a finding. Nguyen et al. (2026) is the only study in this area sweeping several optimisers, and it does so for a different purpose. Until the mechanism is replicated across optimisers and model families, the conservative reading is that it is a property of sampling-based search against a particular class of costs.

An evaluation protocol. Section 3.5 identified three quantities that are cheap to compute and essentially never reported: candidate availability, selection regret against the best available candidate, and proxy-physical rank correlation measured at the first and last search iteration. Standardising these would make the competing diagnoses of Section 3.6 directly comparable for the first time, and is a contribution available independently of any new method.

7 Conclusion

Latent world models for robot planning have converged on a common architecture whose components are studied separately and evaluated jointly, which makes failures hard to attribute. We gave background on how this architecture displaced classical system identification, reviewed the literature along the axes of representation objective and search algorithm, and derived a four-surface taxonomy in which each attribution carries an explicit refutation condition. Applying it exposes a methodological gap: most proposed repairs are evaluated under learned dynamics, where the surface they claim to fix is confounded with ordinary prediction error. A controlled study removing that confound places the binding constraint, in the contact-rich setting tested, on the mapping from representation to cost rather than on prediction—and a series of repairs targeting fidelity leaves planning unchanged.

The through-line is a mismatch of quantities. Representations in this literature are trained to be faithful on distributions chosen by the experimenter; planners consume orderings on distributions chosen by the optimiser. Most of the directions in Section 6 are versions of the question of how to close that gap, and the field currently lacks both the objectives and the evaluation protocol needed to answer it.

References

- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. arXiv:2301.08243.
- Mahmoud Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.
- Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

- Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. *arXiv preprint arXiv:2412.03572*, 2024.
- Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022. arXiv:2105.04906.
- Logan Mondal Bhamidipaty, Esmeralda S. Whitamner, David Abel, and Mykel J. Kochenderfer. Imperfect world models are exploitable. *arXiv preprint arXiv:2605.15960*, 2026.
- Kunlin Cai, Rui Song, Jinghuai Zhang, Kaiyuan Zhang, Pranav Bodapati, Alicia Yu, Fnu Suyu, Mohammad Rostami, Jiaqi Ma, and Yuan Tian. What-if world: A causal benchmark for general world models in embodied scenarios. *arXiv preprint arXiv:2605.27589*, 2026.
- Jiaheng Chen. ATM: Action-consistency transfer matrix for diagnosing and improving latent world models. *arXiv preprint arXiv:2606.09028*, 2026.
- Baoqi Gao, Ruize Han, Miao Wang, and Song Wang. IMWM: Intuition models complement world models for latent planning. *arXiv preprint arXiv:2606.01626*, 2026.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.
- David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems*, 2018. arXiv:1803.10122.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, 2019. arXiv:1811.04551.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 2025. arXiv:2301.04104.
- Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, 2024. arXiv:2310.16828.
- Lukas Helff, Quentin Delfosse, David Steinmann, Ruben Härle, Hikaru Shindo, Patrick Schramowski, Wolfgang Stammer, Kristian Kersting, and Felix Friedrich. LLMs gaming verifiers: RLVR can lead to reward hacking. *arXiv preprint arXiv:2604.15149*, 2026.
- Bohan Hou, Gen Li, Jindou Jia, Tuo An, Xinying Guo, et al. World model for robot learning: A comprehensive survey. *arXiv preprint arXiv:2605.00080*, 2026.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.
- Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. Goodhart’s law in reinforcement learning. In *International Conference on Learning Representations*, 2024.
- Hadi Khalaf, Claudio Mayrink Verdun, Alex Oesterling, Himabindu Lakkaraju, and Flavio du Pin Calmon. Inference-time reward hacking in large language models. *arXiv preprint arXiv:2506.19248*, 2025.

- Rahul Kidambi, Aravind Rajeswaran, Praneeth Netrapalli, and Thorsten Joachims. MOREL: Model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2005.05951.
- Keon Woo Kim. Latent state design for world models under sufficiency constraints. *arXiv preprint arXiv:2605.01694*, 2026.
- Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Learning for Dynamics and Control*, 2020.
- Yann LeCun. A path towards autonomous machine intelligence. Technical report, New York University and Meta AI, 2022. URL <https://openreview.net/forum?id=BZ5a1r-kVsf>. Version 0.9.2, 2022-06-27. Position paper; not peer reviewed.
- Liangyu Li, Shengzhi Wang, and Qingwen Liu. Beyond Euclidean proximity: Repairing latent world models with horizon-matched trajectory reachability metrics. *arXiv preprint arXiv:2605.22164*, 2026a.
- Wenyuan Li, Guang Li, Keisuke Maeda, Takahiro Ogawa, and Miki Haseyama. Predictive but not plannable: RC-aux for latent world models. *arXiv preprint arXiv:2605.07278*, 2026b.
- Yuejiang Liu, Fan Feng, Lingjing Kong, Weifeng Lu, Jinzhou Tang, Kun Zhang, Kevin Murphy, Chelsea Finn, and Yilun Du. World action verifier: Self-improving world models via forward-inverse asymmetry. *arXiv preprint arXiv:2604.01985*, 2026.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorld-Model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.
- Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar, Mido Assran, Koustuv Sinha, and Michael Rabbat. V-JEPA 2.1: Unlocking dense features in video self-supervised learning. *arXiv preprint arXiv:2603.14482*, 2026.
- Heejeong Nam, Quentin Le Lidec, Lucas Maes, Yann LeCun, and Randall Balestriero. Causal-JEPA: Learning world models through object-level latent masking. *arXiv preprint arXiv:2602.11389*, 2026.
- Hoang Nguyen, Xiaohao Xu, and Xiaonan Huang. Latent geometry beyond search: Amortizing planning in world models. *arXiv preprint arXiv:2605.08732*, 2026.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. arXiv:2304.07193.
- Arjun Parthasarathy, Nimit Kalra, Rohun Agrawal, Yann LeCun, Oumayma Bounou, Pavel Izmailov, and Micah Goldblum. Closing the train-test gap in world models for gradient-based planning. *arXiv preprint arXiv:2512.09929*, 2025.
- Cristina Pinneri, Shambhuraj Sawant, Sebastian Blaes, Jan Achterhold, Joerg Stueckler, Michal Rolinek, and Georg Martius. Sample-efficient cross-entropy method for real-time planning. In *Conference on Robot Learning*, 2020. arXiv:2008.06389.
- Reuven Y. Rubinstein. Optimization of computer simulation models with rare events. *European Journal of Operational Research*, 99(1):89–112, 1997.

- Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap, and David Silver. Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588:604–609, 2020. arXiv:1911.08265.
- Gawon Seo, Dongwon Kim, and Suha Kwak. ACID: Action consistency via inverse dynamics for planning with world models. *arXiv preprint arXiv:2607.02403*, 2026.
- Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems*, 2022.
- James E. Smith and Robert L. Winkler. The optimizer’s curse: Skepticism and postdecision surprise in decision analysis. *Management Science*, 52(3):311–322, 2006. doi: 10.1287/mnsc.1050.0451.
- Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from reward-free offline data: A case for planning with latent dynamics models. *arXiv preprint arXiv:2502.14819*, 2025.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *Transactions on Machine Learning Research*, 2026. URL <https://openreview.net/forum?id=cHZn5Gdh8e>.
- Leonardo F. Toso, Davit Shadunts, Yunyang Lu, Nihal Sharma, Donglin Zhan, Nam H. Nguyen, and James Anderson. Learning invariant visual representations for planning with joint-embedding predictive world models. *arXiv preprint arXiv:2602.18639*, 2026.
- Ying Wang, Oumayma Bounou, Gaoyue Zhou, Randall Balestriero, Tim G. J. Rudner, Yann LeCun, and Mengye Ren. Temporal straightening for latent planning. *arXiv preprint arXiv:2603.12231*, 2026.
- Grady Williams, Andrew Aldrich, and Evangelos A. Theodorou. Model predictive path integral control: From theory to parallel computation. *Journal of Guidance, Control, and Dynamics*, 40(2):344–357, 2017. doi: 10.2514/1.G001921.
- Workshop authors. Hidden failure modes in latent world-model planning from offline data. In *ICML Demo Workshop*, 2026.
- Kevin Xia, Yushu Pan, and Elias Bareinboim. Neural causal models for counterfactual identification and estimation. In *International Conference on Learning Representations*, 2023.
- Tianzhuo Yang, Zihan Shen, Zirui Mi, Zhaoyi Zhang, Jiayi Zhou, Jiaming Ji, Juntao Dai, Jiawei Chen, Boyuan Chen, and Yaodong Yang. MiraBench: Evaluating action-conditioned reliability in robotic world models. *arXiv preprint arXiv:2605.29360*, 2026.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020a.
- Tianhe Yu, Garrett Thomas, Lantao Yu, Stefano Ermon, James Y. Zou, Sergey Levine, Chelsea Finn, and Tengyu Ma. MOPO: Model-based offline policy optimization. In *Advances in Neural Information Processing Systems*, 2020b. arXiv:2005.13239.

- Yang Yu, Shiyuan Zhang, Yifei Sheng, Haoxiang Ren, and Haoxin Lin. How should world models be evaluated? a decision-making-centric position. *arXiv preprint arXiv:2606.15032*, 2026.
- Wancong Zhang, Basile Terver, Artem Zholus, Soham Chitnis, Harsh Sutaria, Mido Assran, Randall Balestriero, Amir Bar, Adrien Bardes, Yann LeCun, and Nicolas Ballas. Hierarchical planning with latent world models. *arXiv preprint arXiv:2604.03208*, 2026.
- Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 79115–79135, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>.